UNIT 2 · TOPICS 2.8–2.10 · RANDOM VARIABLES

Random Variables & the Binomial Distribution

Discrete probability distributions, expected value and standard deviation, transforming and combining random variables, and the conditions, formula, mean and standard deviation of a binomial count — with interpretation in context.

1. Discrete Random Variables and Probability Distributions

A **random variable** takes numerical values determined by a random phenomenon. A **discrete** random variable has finitely many (or countable) possible values. Its probability distribution lists every value with its probability; it may be given as a table, a graph or a function, and can be found by the rules of probability or estimated by simulation.

TWO NON-NEGOTIABLE CONDITIONS

$0 \le P(x_i) \le 1$ for every value $\quad \sum P(x_i) = 1$

A **cumulative** probability distribution gives $P(X \le x)$ for each value; the last entry must be exactly 1. Parameters (such as μ_X and σ_X) are fixed numbers describing the distribution — they are not statistics computed from a sample.

x	0	1	2	3
P(X = x)	0.10	0.25	0.40	0.25
P(X ≤ x)	0.10	0.35	0.75	1.00

2. Expected Value, Variance and Standard Deviation

MEAN AND STANDARD DEVIATION OF A DISCRETE RANDOM VARIABLE

$$\mu_X = E(X) = \sum x_i \, P(x_i)$$

$$\sigma^2_X = \mathrm{Var}(X) = \sum (x_i - \mu_X)^2 \, P(x_i)$$

$$\sigma_X = \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$$

Interpretation (required on FRQs): μ_X is the long-run average value of X over many repetitions, in context; σ_X is the typical distance of X from that mean in the long run. For the table above,
$\mu_X = 0(.10) + 1(.25) + 2(.40) + 3(.25) = 1.80$.

3. Linear Transformations and Combining Random Variables

TRANSFORMING X, AND ADDING OR SUBTRACTING INDEPENDENT VARIABLES

$$\mu_{aX+b} = a\mu_X + b \qquad \sigma_{aX+b} = |a|\,\sigma_X$$

$$\mu_{X\pm Y} = \mu_X \pm \mu_Y$$

$$\sigma^2_{X\pm Y} = \sigma^2_X + \sigma^2_Y \quad (X, Y \text{ independent})$$

Adding a constant shifts the centre but **not** the spread; multiplying by a negative constant flips the centre and scales the spread by its absolute value. For a *difference* of independent variables the **variances still add** — never subtract them. Standard deviations are combined by $\sqrt{(\sigma^2_X + \sigma^2_Y)}$, never by $\sigma_X + \sigma_Y$.

4. The Binomial Distribution: Conditions, Probability, Mean and SD

BINS — JUSTIFY BEFORE YOU COMPUTE

$$P(X = k) = (n \text{ choose } k)\, p^k\, (1-p)^{n-k}$$

$$\mu_X = np \qquad \sigma_X = \sqrt{(np(1-p))}$$

Binary: each trial has only success / failure. **I**ndependent trials: when sampling without replacement, check the 10% condition $n \le 0.10N$. **N**umber of trials n is fixed in advance. **S**ame probability of success p on every trial. X counts the number of successes, so $k = 0, 1, \ldots, n$.
$(n \text{ choose } k) = n! \,/\, (k!\,(n-k)!)$.

WORKED EXAMPLE AND CALCULATOR COMMANDS

For $X \sim \text{Bin}(10, 0.30)$: $\mu_X = 3$ successes and $\sigma_X = \sqrt{(10 \cdot .3 \cdot .7)} \approx 1.45$. "Over many sets of 10 trials, we expect about 3 successes, typically varying by about 1.45 from that mean."
`binomPdf(10,.3,4)` gives $P(X = 4)$; `binomCdf(10,.3,4)` gives $P(X \le 4)$; `randBin(10,.3,500)` simulates 500 such counts.

5. Exam Traps

TRAPS THE EXAM ACTUALLY TESTS

- **State the BINS conditions in context** before using the formula; "independent" needs the 10% condition when sampling without replacement.
- **binomPdf vs binomCdf:** exactly k needs Pdf; "at most k" needs Cdf; "at least k" is $1 - \text{binomCdf}(n, p, k - 1)$.
- **Variance is not additive in SD:** add variances, then take the square root.
- **A difference subtracts means but ADDS variances** for independent variables.
- **Interpret μ and σ in context** with "long run" / "typical", not "the most likely value".
- **$P(X = x) = 0$ for continuous variables** — only intervals have probability (see the Normal distribution sheet).